

PREDICTIVE MODELLING REPORT

Customer Churn Analysis for a Telecommunications Company



October 25, 2026

Table of Contents

1. OVERVIEW **2**

2. ANN MODEL CONFIGURATION **2**

3. MODEL PERFORMANCE **2**

3.1 CONFUSION MATRIX SUMMARY 2

4. KEY FINDINGS **3**

5. BUSINESS INSIGHT **3**

6. MAIN RECOMMENDATIONS **3**

7. PREDICTIVE MODEL VISUALISATIONS **3**

8. OVERALL CONCLUSION **6**

REFERENCES **7**

1. Overview

There is an outcome variable (churn) that is binary; 12 predictor variables were encoded (continuous data); and 6,741 customer records are used in the analysis. 26.57% of customers churn, with a number of churn cases being 1791. All earlier published cluster and PCA columns are purely for descriptive purposes and not used as model inputs. The source repository is provided by Goreja (n.d.) and the validation checks by Pandas Development Team (n.d.).

ANN, logistic regression and random forest are compared in this workflow. A new four-segment K-Means feature is trained solely based on the data acquired from the development records. Validation chooses models and thresholds and ensures independence of the held out test data (Scikit-learn Developers, n.d.).

2. ANN Model Configuration

The ANN has 16 inputs, of which 12 are for using as source predictors and the remaining four are for training only, and serve as cluster indicators. Dense layers with dropouts are used and feed into a sigmoid churn output. Training is done with Adam and binary cross-entropy, as detailed by the Keras Team (n.d.).

Table 1. ANN model configuration

Component	Implemented configuration
Input representation	12 source predictors and four training-only cluster indicators
Network	Dense layers with dropout and sigmoid output
Optimisation	Adam and binary cross-entropy
Training control	Early stopping, maximum 45 epochs, 21 completed
Model comparison	ANN, logistic regression, random forest
Split	4,313 train (64%), 1,079 validation (16%), 1,349 test (20%)
Selection	Highest validation average precision
Threshold	F2-maximising validation threshold, ANN ≈ 0.298

3. Model Performance

Test ROC-AUC and PR-AUC values for the ANN were 0.810 and 0.587, respectively. At the validation-selected threshold of 0.298, it reached 91.90% recall, 39.98% precision, 55.72% F1 and 61.23% accuracy. The ANN outperformed logistic regression slightly on the test measures, despite the latter outperforming the former on the validation average precision.

Table 2. Held-out model performance at validation-selected thresholds

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
ANN	61.23%	39.98%	91.90%	55.72%	0.810	0.587
Logistic regression	61.53%	40.24%	92.74%	56.13%	0.817	0.596
Random forest	57.45%	37.47%	90.22%	52.95%	0.789	0.542

3.1 Confusion Matrix Summary

The ANN correctly classifies 329, and incorrectly classifies 29 test churners. It also identified 494 customers that continued whilst successfully identifying 497 customers. These false alerts are important as they eat up resources and can impact on customer experience.

Table 3. ANN confusion matrix outcomes

Outcome	Customer count	Business interpretation
True negative	497	Correctly predicted to remain
False positive	494	Unnecessary churn-risk alert

False negative	29	Actual churner not identified
True positive	329	Actual churner identified

4. Key Findings

The threshold is used to detect churn while providing lesser accuracy. At the default 0.50 cutoff, the ANN achieved 71.98% accuracy, 75.70% recall and 58.91% F1. Early stopping caused training to be stopped at epoch 21. Call cut-off should be based on the contact capacity and customers value.

5. Business Insight

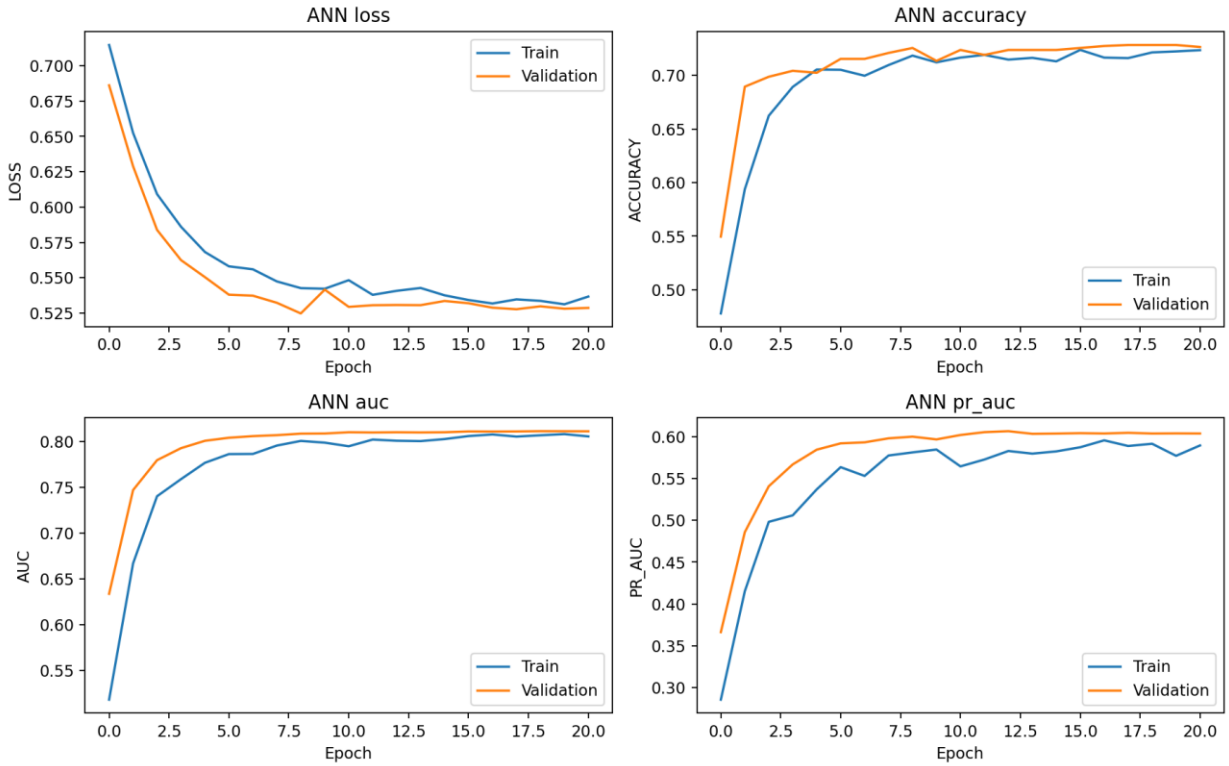
Churn probability is used for a ranking signal and not to show that cancellation must be done. It can be used for selective outreach and the observed Churn rate in Cluster 3 is 48.84%. Supervised predictors do not include clusters published in full-dataset.

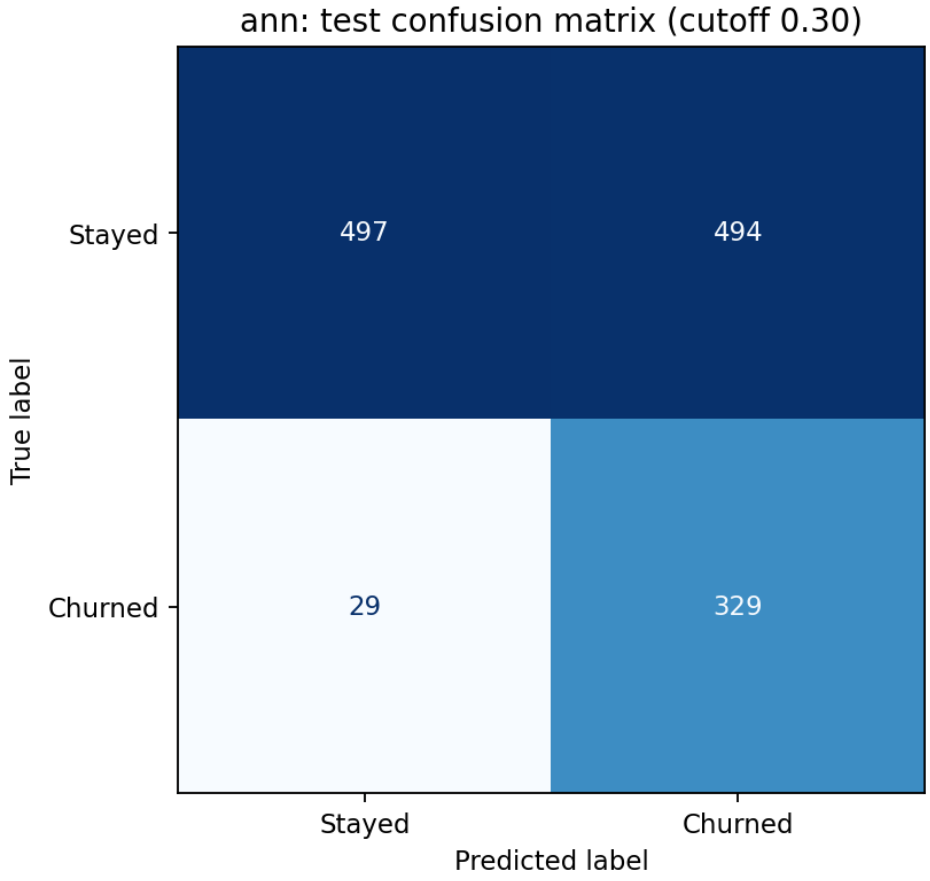
6. Main Recommendations

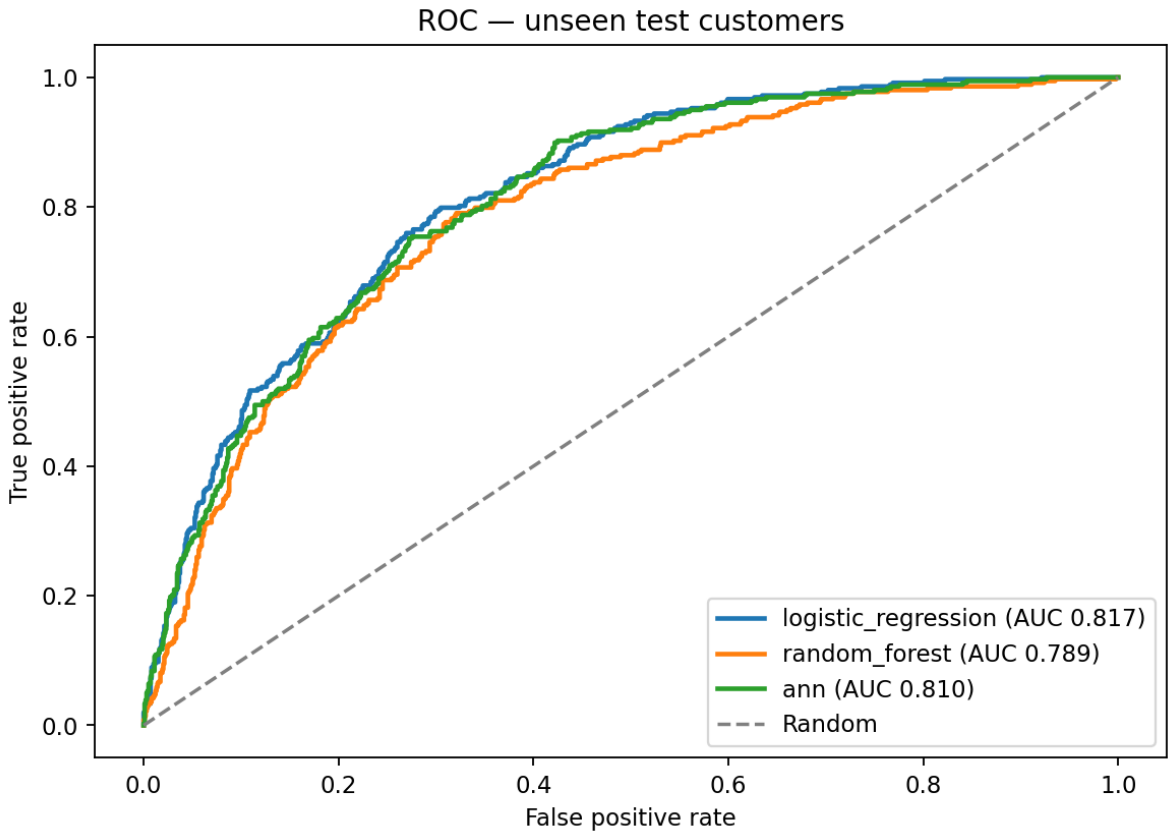
The controlled retention pilot should focus on the highest risk customers who have value to your accounts. The decision on the selected threshold should be guided by the following - offer costs, contact capacity and a true comparison group to decide the boundaries that accrue net retention benefits. Preprocessing and independent test has to be preserved in subsequent retraining.

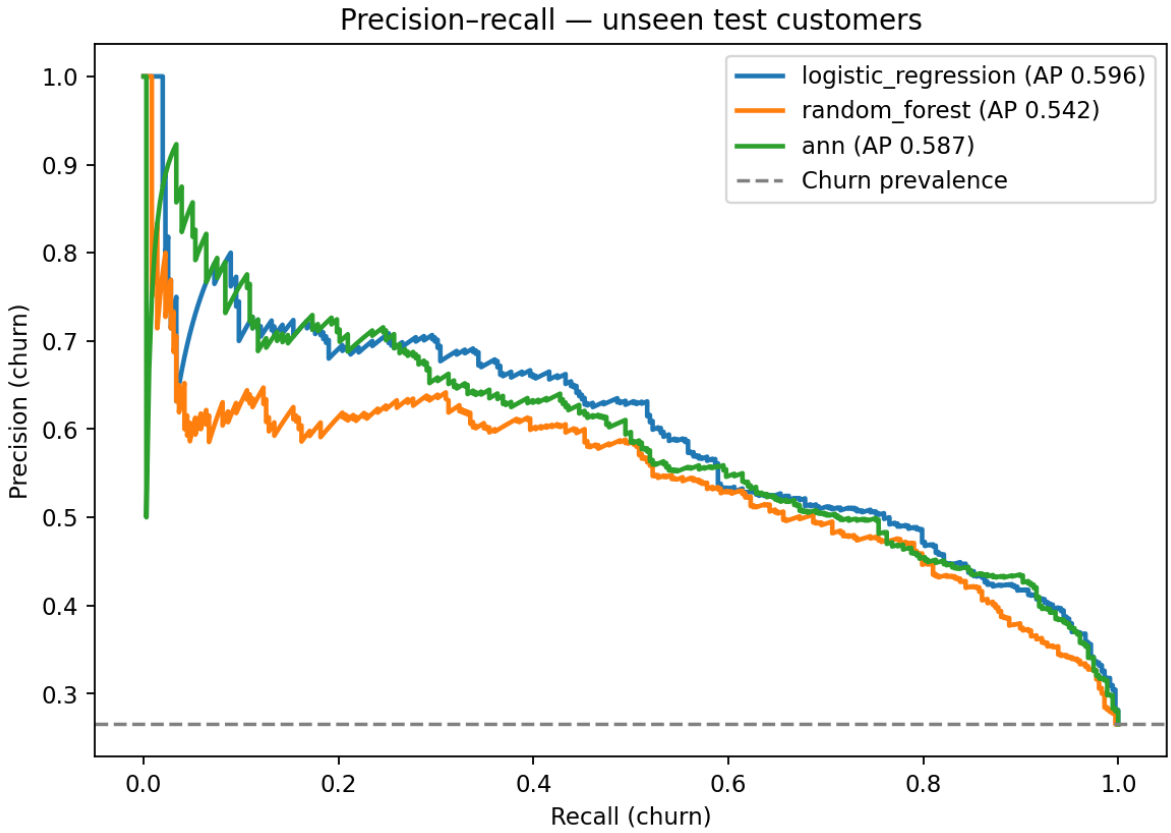
7. Predictive Model Visualisations

The plots show the completed ANN learning history, confusion matrix and ROC and precision–recall comparisons. They explain what the convergence of training is and what impact the lowered decision threshold (Matplotlib Development Team, n.d.) has. The hosted notebook environment is documented in Google (n.d.).




ann: test confusion matrix (cutoff 0.30)
Stayed
Churned
True label
497
494
29
329
Stayed
Churned
Predicted label


ROC — unseen test customers
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
logistic_regression (AUC 0.817)
random_forest (AUC 0.789)
ann (AUC 0.810)
Random


Precision–recall — unseen test customers
logistic_regression (AP 0.596)
random_forest (AP 0.542)
ann (AP 0.587)
Churn prevalence
Precision (churn)
Recall (churn)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.0
0.2
0.4
0.6
0.8
1.0

8. Overall Conclusion

The ANN is a prototype to rank churn and to achieve a high recall, but is not a proven solution for retention. Test comparison demonstrates that in some cases a more complex network isn't always better. Validated transformations, cost based threshold selection and monitoring are also required for deployment.

References

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